COM209 Engineering Mathematics V

Tutorial 6

1. Let X and Y be two independent standard uniform random variables. Find

   (a) $E[XY]$

   (b) $E[e^{X+Y}]$

   (c) $E[X^2 + Y^2 + XY]$

   (d) $E[Ye^{XY}]$

2. Discussion of problem 7.4:

7.4. If X and Y have joint density function

$$f_{X,Y}(x,y) = \begin{cases} 1/y, & \text{if } 0 < y < 1, \ 0 < x < y \\ 0, & \text{otherwise} \end{cases}$$

find

(a) $E[XY]$

(b) $E[X]$

(c) $E[Y]$

3. Discussion of problem 7.30:

7.30. If X and Y are independent and identically distributed with mean μ and variance σ^2, find

$$E[(X - Y)^2]$$

4. Discussion of problems 7.40 and 7.50:

7.40. The joint density function of X and Y is given by

$$f(x,y) = \frac{1}{y}e^{-(y+x/y)}, \quad x > 0, y > 0$$

Find $E[X]$, $E[Y]$, and show that $\text{Cov}(X,Y) = 1$.

7.50. The joint density of X and Y is given by

$$f(x,y) = \frac{e^{-x/y}e^{-y}}{y}, \quad 0 < x < \infty, \quad 0 < y < \infty$$

Compute $E[X^2|Y = y]$.

5. An instructor has 50 exams that will be graded in sequence. The times required to grade the 50 exams are independent, with a common distribution that has mean 20 minutes and standard deviation 4 minutes. Approximate the probability that the instructor will grade at least 25 of the exams in the first 450 minutes of work.

6. Engineers believe that W, the amount of weight (in units of 1000 pounds) that a certain span of a bridge can withstand without structural damage resulting, is normally distributed with mean 400 and standard deviation 40. Suppose that the weight (again, in units of 1000 pounds) of a car is a random variable with mean 3 and standard deviation 0.3. Approximately how many cars would have to be on the bridge span for the probability of structural damage to exceed 0.1?